pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Course 2 Capstone

WEEK 3 · DAY 5

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The brief: the desk lead's note and the cover note

The Desk Lead's Note

Month end. Operations sent November's extract with a cover note. Before any number leaves the desk, prove the file: every column repriced with your own "bondmath", every difference explained, the fixes in one function so next month is one call. Then the risk report as a workbook: DV01, duration, spread and DTS by sector and rating, and the rate and spread what-ifs. Run the same functions on September and show me what changed. One page. Describe it: no views on any bond, issuer, or sector.

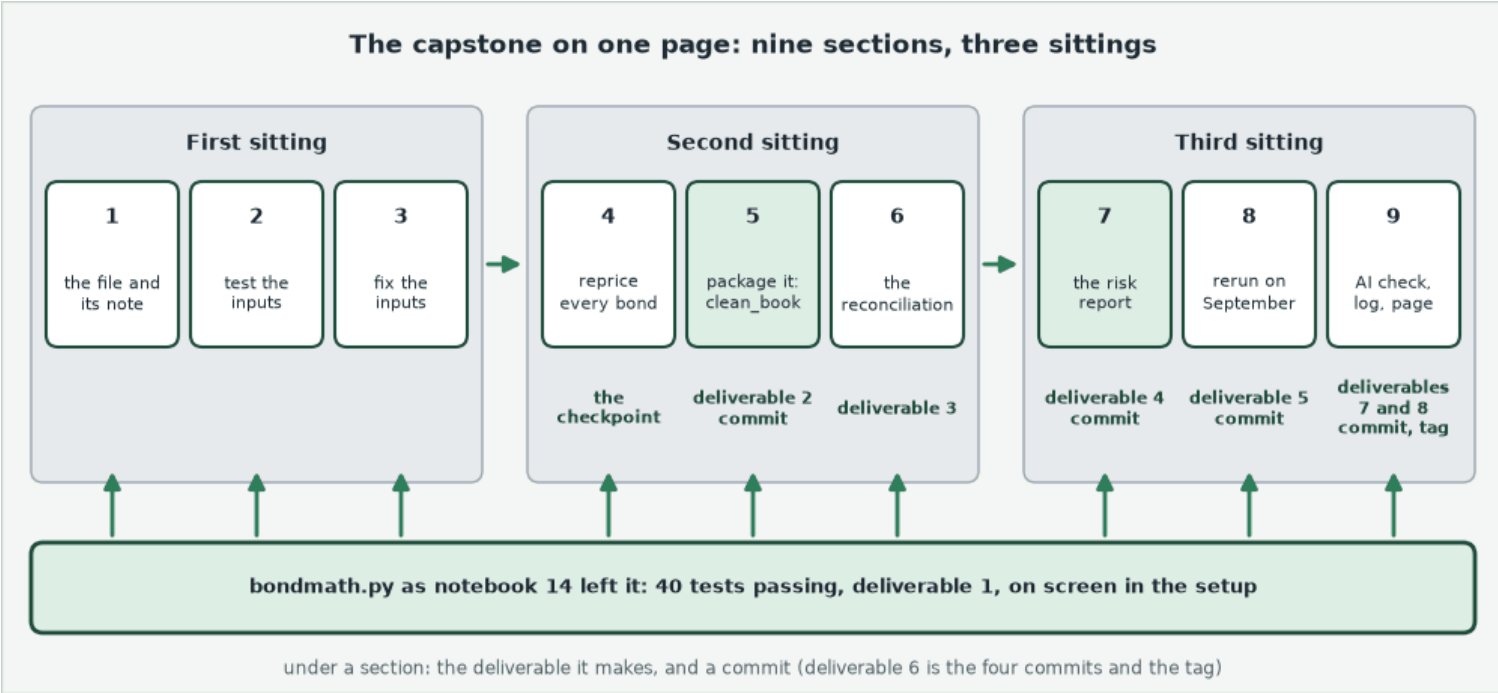
The Cover Note

```
To: Credit desk
From: Operations
Subject: Month-end holdings extract, as of 2026-11-30
The month-end extract is attached as course2_capstone_holdings.csv. It is the same layout as last month.
Control totals for the book:
  Number of bonds:      199
  Total face:           493,000,000 dollars
  Total market value:   496,372,835.00 dollars
Market value is face times clean price plus accrued interest, divided by 100.
All positions settle and are priced as of 2026-11-30. Analytics are semiannual, 30/360.
Please tell us if the extract does not agree with these totals.
```

The file's problems are planted and unnamed, and every name is invented.

Before any number leaves the desk, prove the file.

The capstone on one page: nine sections, three sittings



Nine sections, three sittings, one page at the end. Your module does the checking.

The seven rules, and how it is checked

The Rules

1. Work in a copy: fixes go to "book", never to "raw".
2. State every decision in a markdown cell: what, how many rows, what evidence.

3. Print "headline()" before and after every fix.
4. Every analytic comes from "bondmath", on the cover note's conventions.
5. Describe: never recommend, rank, or forecast; a scenario is a what-if.
6. Every Git command names the work folder: "!git -C "{work}" ...".
7. Restart and Run All before you hand in.

An AI assistant is allowed: the docstring is the prompt, and every answer is tested and logged in 9.2.

How It Is Checked

Each "check()" compares a fingerprint of your answer with a stored one; round as each question asks. The checkpoint after section 4 must pass before section 5; a Stuck block holds the decision it expects: open it after your own. Worked answers: "course2_capstone_solutions.ipynb", once your page is written.

The data, and the setup

Units: "coupon" and "yield_pct" in percent, "oas" in basis points, "duration" and "spread_duration" in years, "dts" in years times basis points, "price" and "accrued" per 100 of face, and "par_held", "amount_outstanding", and "market_value" in dollars.

File	What it is
course2_capstone_holdings.csv	November's extract, in the 19 columns Course 1's project pictured
course2_capstone_cover_note.txt	the cover note above
benchmark.csv	821 bonds as of 2026-09-30
course2_treasury_curve.csv	the desk's curve file (notebook 11)
ratings.csv	the rating scale
holdings.csv	September's book, clean (notebooks 06 to 14)

"oas" is the spread to notebook 12's formula curve, and "dts" is "spread_duration" times "oas", to 1 decimal.

Setup

Run these cells first and do not change them.

Your own "my_bondmath" may stand in: copy its two files in, your eight characters in place of "k4w7q2xr", then rerun the next cell.

```
Copy-Item "$HOME\projects\my_bondmath\bondmath.py", "$HOME\projects\my_bondmath\test_bondmath.py" "$env:TEMP\pfi_course2_15_k4w7q2xr"
```

Deliverable 1 is on screen: 40 tests pass.

1 The file and its note; 2 seven rules

1.1 Rows, Columns, and a Working Copy

What shape is the extract (answer_1_1)? Copy "raw" to "book": every fix from here goes to "book".

1.2 The File Less Its Note

Print "headline()" as received, then the file less the note in rows, face (answer_1_2), and market value. What do the gaps suggest?

The totals test a file first; section 2 finds what moves them.

The note's totals count dollars; seven rules test the rest. Count on "book" as 1.1 made it, and fix nothing until section 3.

Rule	What the desk expects
1	one row for each bond, no row a copy
2	the maturity after the as-of date
3	no number missing
4	coupon and yield in percent, price and accrued per 100 of face
5	market value is face x (price + accrued) / 100, to the cent
6	every analytic reproduces from the bond's terms with bondmath (section 4)
7	CUSIPs, ratings, sizes, spread durations, and price dates well formed

Tools: Course 1's "duplicated()", "isna()", "describe()", ".str.fullmatch()", "isin()". Text dates written year-month-day sort in date order, so "<=" compares them.

Every rule gets a count: failures are the fix list, passes go in the page.

2 Test the inputs: count first, fix nothing yet

2.1 Rule 1: One Row for Each Bond

How many rows copy an earlier row exactly (answer_2_1)? Show each beside its original, with the face and value they carry.

2.2 Rule 2: The Maturity

How many rows mature on or before the as-of date (answer_2_2)? Show them with coupon, maturity, and duration: which column is wrong?

2.3 Rule 3: Missing Numbers

How many cells are missing (answer_2_3)? Which of each row's other numbers gives the blank back through your module?

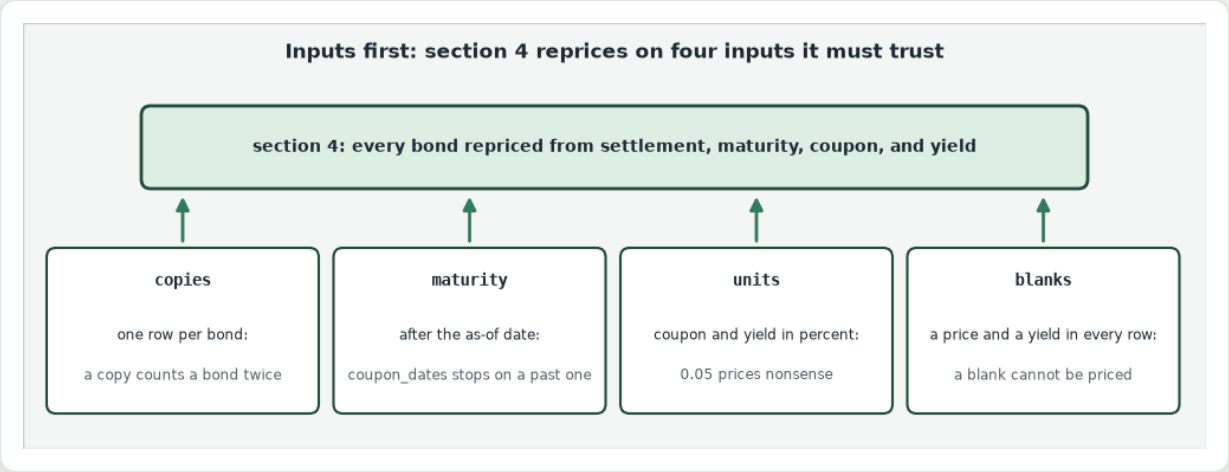
2.4 Rule 4: Units

Summarize "coupon" and "yield_pct". How many coupons (answer_2_4a) and yields (answer_2_4b) are below 1 percent? Show those rows with price and accrued.

2.5 Rules 5 and 7

How many rows miss rule 5 by more than half a cent (answer_2_5a), and how many could be tested? Total the breaks of rule 7 (answer_2_5b).

3 Fix the inputs: four inputs, each with a decision



Each fix gets a decision cell, then "headline()" before and after it.

Tools: Course 1's "drop_duplicates()", ".loc[mask, column] ="; "date.fromisoformat" (notebook 06).

3.1 Copies, Maturity, and Units

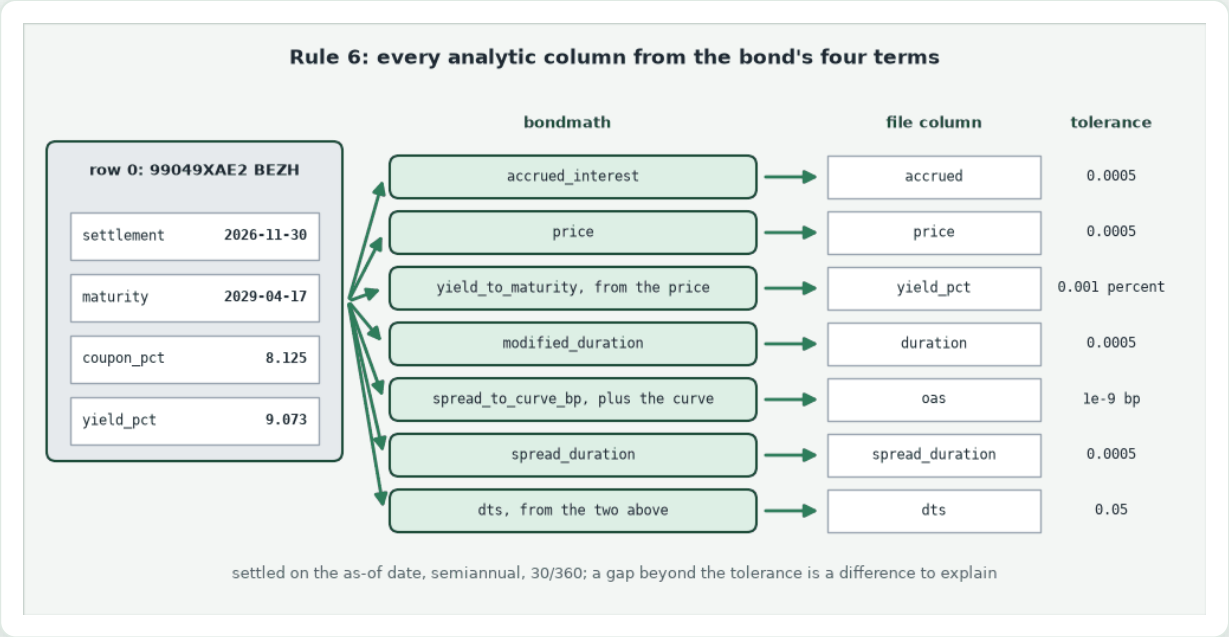
Make your three fixes. Give the face after them (answer_3_1a), the corrected maturity as text year-month-day (answer_3_1b), and the mean coupon to 3 places (answer_3_1c).

3.2 Fill the Blanks

Fill each blank from the other number of its pair. Give the sum of the yields you filled (answer_3_2a) and of the prices (answer_3_2b), each to 3 places.

Totals that tie cannot see a wrong price; section 4 can. The first sitting ends here.

4 Reprice every bond from four terms



The second sitting: recompute each column into a "_calc" column, count the rows beyond notebook 10's tolerance, then decide and fix.

Tools: "ROUNDED_3" (notebook 10), "*bond" (notebook 08), ".dt.is_month_end" (notebook 06).

4.1 Accrued Interest

Add "accrued_calc". How many rows of "accrued" are beyond tolerance (answer_4_1)? Set each beside coupon x actual days since the last coupon / 360. Decide and fix.

4.2 Price and Yield

Add "price_calc". How many prices are beyond tolerance (answer_4_2a)? Compare each with "price_calc" plus accrued; decide and fix. Then give the largest gap of "yield_calc", from the price, to "yield_pct", to 4 places (answer_4_2b).

4.3 Duration

Add "duration_calc". How many durations are beyond tolerance (answer_4_3)? Set each beside "spread_duration" and "macaulay_duration". Decide and fix.

The special bonds, DV01, spread, DTS, checkpoint

One bond matures on a month end (notebook 06's rule); one has a single coupon left (notebook 08).

4.4 The Two Special Bonds

Print the month-end bond's "coupon_dates", 30/360 days accrued, and accrued, and give its next coupon date as text (answer_4_4a). Print the final-period bond's A, E, and DSC, and give its dirty price to 6 places (answer_4_4b).

Excel side by side. Tested in desktop Excel for Windows.

```
Month-end bond, next coupon
=COUPNCD(DATE(2026,11,30),DATE(2033,8,31),2,0)
Its clean price
=PRICE(DATE(2026,11,30),DATE(2033,8,31),5.5/100,5.7/100,100,2,0)
Final-period bond, clean price
=PRICE(DATE(2026,11,30),DATE(2027,3,15),4.25/100,4.193/100,100,2,0)
```

4.5 Dirty Price, DV01, and Convexity

Add "dirty_price", "dv01", and "convexity", and check each DV01 against notebook 09's bump, leaving out the final-period bond. Give the book's DV01 in dollars, to 2 places (answer_4_5).

4.6 Spread to the Curve, Two Ways

Add "spread_bp" on notebook 12's formula curve, and "spread_bp_table" on "curve". How many "spread_bp" are more than 1e-9 bp from "oas" (answer_4_6a)? Give the largest gap of "spread_bp_table" to "oas", to 2 places (answer_4_6b).

4.7 Spread Duration and DTS

Add "spread_duration_calc", and "dts_calc" from it and "spread_bp". How many "spread_duration" (answer_4_7a) and "dts" (answer_4_7b, tolerance 0.05 + 1e-9) are beyond tolerance? Recompute DTS from the file's own "spread_duration": what explains them?

Checkpoint: The Repriced Book

Does "book" match the cover note and the expected decisions? Every line must read "correct" before section 5.

5 Package it as one call; 6 the reconciliation

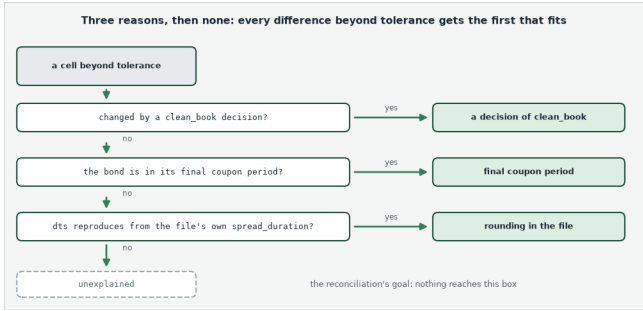
Every column reprices, or has a reason it does not: now one call does it all.

5.1 "clean_book", Docstring First

In "capstone.py", written with "%%writefile" in a cell of its own, write "clean_book(raw, benchmark, curve)": a docstring listing your seven decisions, then the decisions and section 4's columns, its inputs unchanged. "assert" each rule it promises, with a message. Store its result on "raw" in cleaned, "assert" the control totals, and show "cleaned" equals "book".

5.2 Commit It

Commit "capstone.py" alone, with a subject naming the deliverable, and show the history.



A reconciliation sets the file's numbers beside your module's and gives every difference a reason.

6.1 Delivered Against Recomputed

Line up the extract, without its copies, row for row with "cleaned". Count the cells beyond tolerance (answer_6_1): "coupon" and "maturity" against the cleaned values, and the eight other analytic and value columns against what "bondmath" recomputed. A blank is a difference.

6.2 A Reason for Every Difference

Build reconciliation, one row per difference: CUSIP, column, delivered, recomputed, and the first reason in the picture's order. How many are unexplained (answer_6_2a), and how many bonds did a decision change (answer_6_2b)?

7 The risk report, the what-ifs, a workbook

The third sitting, on notebook 14's risk card and full reprice.

7.1 risk_report

Append "risk_report(book, ratings)" to "capstone.py" with "%%writefile -a". It returns a dict of three tables:

"positions": one row per bond, adding bucket, grade, "weight_pct", "dv01_dollars", and "dts_contribution";

"summary": the book, each sector, and each bucket in credit order: bonds, value, weight, DV01 in dollars, and value-weighted duration, spread, and DTS;

"scenarios": rates -100, -25, +25, +100 bp, spreads +10 percent and +25 bp, repriced in full and estimated, in dollars.

Store its result in report, and the book's duration to 3 places (answer_7_1).

7.2 By Sector and Rating

Which sector carries the largest share of the book's DTS (answer_7_2a)? What percent of the DTS is High Yield, to 2 places (answer_7_2b)? Set each beside its share of value.

7.3 The What-Ifs

Give the +100 bp full reprice in dollars (answer_7_3a), and the -100 bp estimate less full reprice (answer_7_3b), both to 2 places. Why do the two part for a large move? Each is a mechanical what-if, never a forecast.

7.4 The Workbook

Write "course2_risk_report.xlsx", a sheet per table, no row labels. Read it back and "assert" the positions sheet adds to the book's DV01 and the note's value. Give the sheet names as a list (answer_7_4a) and the positions rows (answer_7_4b).

```
Book DV01 in dollars
=SUM(R2:R200)
Book duration
=SUMPRODUCT(N2:N200,P2:P200)/SUM(N2:N200)
Book DTS
=SUMPRODUCT(N2:N200,V2:V200)/SUM(N2:N200)
```

7.5 Commit the Code, Not the Workbook

Read "git status --short" and "git diff --stat", then commit "capstone.py" alone, leaving the workbook untracked (notebook 14).

8 Rerun on September, do not redo

8.1 September Through Both Functions

Run "clean_book", then "risk_report", on "september", unchanged. Do its delivered columns come back untouched? Give September's book DV01 in dollars, to 2 places (answer_8_1).

8.2 A Test for Each Promise

Write "test_capstone.py" with "test_clean_book_ties_to_the_cover_note" (November's three totals) and "test_clean_book_leaves_september_alone". Run pytest below, give the count passed (answer_8_2), and commit the test file.

8.3 Two Months Side by Side

Build comparison, columns "September", "November", and "change", with a row each for bonds, market value, DV01, duration, spread, and DTS. Give the DTS change, November less September, to 1 place (answer_8_3).

The Change in Percent

Which figures moved most from September? This cell is written for you.

Two months, one call each, and a test that says so.

9 The AI check, the log, the page, and the tag

You asked an assistant for one step of "clean_book", with the docstring as the prompt. This came back, written for this course in the style of an assistant's answer: it runs, and it breaks one convention.

```
def recompute_accrued(book):
    """Accrued interest per 100 of face for every bond in a month-end book.
    book has as_of_date and maturity (text, year-month-day) and coupon (percent a year).
    Settlement is each row's as_of_date. Convention: the book's own, as the cover note states:
    semiannual coupons on US 30/360. Returns a pandas Series of floats in the book's order.
    """
    settlements = [date.fromisoformat(text) for text in book['as_of_date']]
    maturities = [date.fromisoformat(text) for text in book['maturity']]
    # count the actual days in the period, the most precise day count
    accrued = [bondmath.accrued_interest(s, m, c, 2, 'ACT/ACT')
               for s, m, c in zip(settlements, maturities, book['coupon'])]
    return pd.Series(accrued, index=book.index)
```

What should it return for the month-end bond? Your 4.4 printed it. Write down a number.

Test it against Excel's accrued on the reference file's 200 holdings rows before you trust it.

Find the convention it broke, fix the function, and rerun until the test passes. On "cleaned", give your function's accrued for the month-end bond, to 6 places (ai_month_end), and the bonds the assistant's version gets wrong by more than 1e-9 (ai_rows_changed).

9.2 The AI-Use Log

Write one row per question you asked an AI assistant; "none used" is valid. Run the cell to write "ai_use_log.md".

Nothing from an employer, no client names, positions, or credentials, goes into an AI tool or Colab; your firm's policy comes first.

9.3 The Page

Write the page for the portfolio manager, 300 to 450 words, in five parts:

- the book in numbers, with units;
- the risk by sector and rating;
- each what-if, full reprice against estimate;
- September against November;
- the data: each problem, its rows, fix, and evidence, and what was clean.

Describe, never recommend.

Commit, Tag, and Count

Commit the log and the page together, tag the commit "course2", and show "git log --oneline --decorate". How many commits are there (answer_git)?

Bring it home, and before you hand in

Copy the capstone, the workbook, and the four files "test_capstone.py" reads into your own "my_bondmath"; none of yours is replaced. In PowerShell, with the course's ".venv" active and your eight characters in place of "k4w7q2xr":

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_15_k4w7q2xr"
Copy-Item "$work\capstone.py", "$work\test_capstone.py", "$work\course2_risk_report.xlsx" .
Copy-Item "$work\course2_capstone_holdings.csv", "$work\holdings.csv" .
Copy-Item "$work\benchmark.csv", "$work\course2_treasury_curve.csv" .
python -m pytest -q
git add capstone.py test_capstone.py
git commit -m "Course 2 capstone: clean_book, risk_report, and their tests"
git tag course2
```

The workbook stays untracked, to open in Excel; in Colab, download the files from the file browser. The work folder stays in your temp folder.

Before You Hand In

Hand in this notebook, run from the top, with:

- every check "correct" and "raw" unchanged;
- a decision cell and headlines for every fix;
- both functions with docstrings and "assert"s, and "test_capstone.py" passing;
- four commits, the tag, the log, and the page.

The planted problems are in the repo's "data/course2_capstone_answer_key.md": read it after you finish.

The note, and the course

The note asked	Where
Prove the file, every difference explained	sections 2 to 4, 6
The fixes in one function	clean_book
The risk report as a workbook	risk_report, 7.4
September against November	section 8
One page	9.3

In notebook 01 your module was 2 functions and 3 tests; today it is 25 and 40, with Excel as the answer key.

Since then you have learned to:

price a bond between coupon dates, clean and dirty, to Excel's last digit;

measure its duration, DV01, convexity, spread, and DTS;

add up a book and reprice it under what-ifs;

keep every function tested and every change in Git;

prove a file against its note before a number leaves the desk.

Be proud: you took a file nobody vouched for and handed back a risk report you can defend line by line.

Next is Course 3, Market Data and Time Series: FRED series turned into changes, volatility, and curve shapes in "marketdata.py", beside "bondmath.py".